Prof Elective 5 — Advanced Web Development: Back End

Worksheet 4.2 — Database Failure and Recovery Drill

Guided Laboratory Activity | Self-paced / Asynchronous Learning

Objective

Demonstrate the difference between transaction rollback and backup restoration, create and restore a database backup in a safe test environment, and verify that recovered data is usable and consistent.

Expected Output

A short recovery report containing a test database backup, a restored database, verification evidence, screenshots, and answers explaining which recovery method applies to different failure scenarios.

SAFETY FIRST
Perform every step on your own local development database or another authorized test database. Never delete data, stop services, or test recovery on a live university database. Do not use real student personal data for this activity.

Before You Start

You will use the MySQL database from your existing Laravel Books project. The exact database name may differ in your project. Check the DB_DATABASE value in your .env file, but do not submit or share database passwords.

Tools: Laravel project, MySQL/phpMyAdmin, and a text editor. A terminal is required for the optional command-line backup method. The phpMyAdmin method is sufficient if command-line access is unavailable.

A. Understand the Two Recovery Situations

Situation	What happened?	Appropriate response
Situation 1	A transaction has made changes but has not committed, and an operation fails.	Roll back the transaction, provided the changes use a transactional storage engine.
Situation 2	A record was deleted and the deletion was committed.	Restore from an appropriate backup and, if configured, apply retained logs. A normal rollback cannot undo an already committed deletion.

Answer before proceeding:

1. What is the difference between rolling back an uncommitted transaction and restoring a database backup?
2. Why is a backup not considered reliable until restoration has been tested?

Response to Question 1:

_____

_____

Response to Question 2:

_____

_____

B. Check the Test Database

3. Open your Laravel Books project.
4. Open the .env file and identify DB_DATABASE. Do not copy DB_PASSWORD into your report.
5. Open phpMyAdmin and select that database.
6. Locate the books table and confirm that it contains records.
7. Record the current number of rows in the books table. In phpMyAdmin, use the table's Browse view or run the read-only query below.

```
SELECT COUNT(*) AS total_books FROM books;
```

Write the database name used in your local environment and the initial number of books. Do not include credentials.

Test database name:

_____

_____

Initial number of books:

_____

_____

CHECKPOINT 1
Confirm that this is a local or authorized test database. If you are uncertain whether the database is live, stop and ask your instructor before continuing.

C. Demonstrate Transaction Rollback

This part demonstrates what happens to uncommitted changes. Use a harmless temporary table so the exercise does not change your existing books data.

1. Create a temporary recovery-test table

In phpMyAdmin, select your test database, open the SQL tab, and execute:

```
CREATE TABLE recovery_lab_test (
    id INT AUTO_INCREMENT PRIMARY KEY,
    note VARCHAR(100) NOT NULL
) ENGINE=InnoDB;
```

If the table already exists from an earlier attempt, inspect it first. Do not delete tables unless you are certain they are only your own lab table.

2. Start a transaction and insert a test record

Execute the following as one SQL batch, using the same database connection:

```
START TRANSACTION;

INSERT INTO recovery_lab_test (note)
VALUES ('This row should be rolled back');

SELECT * FROM recovery_lab_test;

ROLLBACK;
```

```
SELECT * FROM recovery_lab_test;
```

Expected result: the first SELECT, before ROLLBACK, shows the temporary row. The second SELECT, after ROLLBACK, should not show that inserted row.

If your phpMyAdmin setup does not keep all statements in the same connection, run START TRANSACTION, INSERT, and ROLLBACK in the same SQL session. Do not assume a transaction spans separate connections.

3. Record your observations

What did the first SELECT show?:

_____

_____

What did the second SELECT show?:

_____

_____

Take Screenshot 1 showing the SQL statements and the result after rollback, if your interface can display it clearly.

D. Create a Backup of the Test Database

Choose ONE of the following methods. The phpMyAdmin method is recommended for students who do not have terminal access.

Method 1 — phpMyAdmin

8. In phpMyAdmin, select the test database used by your Laravel project.
9. Click Export.
10. Choose the Quick export method and SQL format for this exercise.
11. Click Export and save the downloaded .sql file in a folder named Worksheet_4.2_Backup.
12. Rename the file clearly, for example books_lab_backup.sql.

Method 2 — MySQL command line (optional)

Use this method only if the MySQL client utilities are installed and you know how to connect to your local test database. Replace the placeholders with your own local settings. Do not place passwords in screenshots or submit them.

```
mysqldump -u YOUR_USERNAME -p YOUR_TEST_DATABASE > books_lab_backup.sql
```

Enter the password only at the prompt. Confirm that the backup file exists and is not empty. A successful export is not proof that restoration will succeed; the next part tests that.

Take Screenshot 2 showing the exported backup file. Hide any credentials or sensitive data.

E. Restore the Backup to a Separate Test Database

Restore into a new database so that you do not overwrite the database used by your Laravel application.

1. Create a separate destination database

In phpMyAdmin, create a new empty database named books_recovery_test. If you do not have permission to create databases, ask your instructor for an approved destination database. Do not restore over the original database.

2. Import the SQL backup

13. Select books_recovery_test in phpMyAdmin.

14. Click Import.
15. Choose the books_lab_backup.sql file you exported.
16. Keep the SQL format and start the import.
17. Wait for the success message. If an error appears, record it and investigate before continuing.

3. Verify the restored data

Run these read-only queries against books_recovery_test:

```
SELECT COUNT(*) AS total_books FROM books;

SELECT id, title, stock, amount
FROM books
ORDER BY id
LIMIT 10;
```

Compare the row count and sample records with the original database. If the data changed between export and verification, explain the difference rather than claiming the counts must match.

Restored number of books:

_____

_____

Did the sample records match? Explain.:

_____

_____

Take Screenshot 3 showing the destination database and verification query results.

IMPORTANT
Do not change Laravel's .env to point to the recovery database for this exercise. The goal is to inspect the restored copy without disrupting your working application.

F. Failure Scenario Analysis

For each scenario, select the most appropriate recovery approach and explain why.

Scenario	Your proposed response
1. An insert fails before a transaction commits.	
2. An administrator deletes records and commits the deletion.	
3. The database server crashes unexpectedly.	
4. A backup file exists, but no one has tested importing it.	
5. A server's storage is permanently lost.	

Consider transaction rollback, database crash recovery, backups, retained logs/point-in-time recovery, and restoration to replacement infrastructure. The best answer depends on what was configured before the failure.

G. Recovery Objectives: RPO and RTO

RPO (Recovery Point Objective) is the maximum acceptable data-loss interval measured in time. RTO (Recovery Time Objective) is the target time to restore service after a disruption.

Scenario: A university sets an RPO of 15 minutes and an RTO of two hours for its enrollment system.

18. Would a daily backup alone normally demonstrate that the 15-minute RPO can be met? Explain.
19. What additional logging, backup, infrastructure, or testing arrangements might be needed?
20. How would you test whether the two-hour RTO is realistic?

Answer: _____
_____

Answer: _____
_____

Answer: _____
_____

H. Final Reflection

21. Why is transaction rollback not a replacement for database backups?
22. Why should a recovery backup be restored to a separate test database before it is used to replace a production database?
23. What evidence from this activity gives you confidence that your backup is usable? What limitations remain?
24. What would you recommend a small university do to reduce the risk of losing enrollment or grade records?

Answer: _____
_____

Answer: _____
_____

Answer: _____
_____

Answer: _____
_____

I. Submission Requirements

- Screenshot 1: Transaction rollback demonstration.
- Screenshot 2: Backup file created and saved.
- Screenshot 3: Restored database with verification queries.
- Answers to Parts A, C, E, F, G, and H.
- Short report stating the backup method, restore method, verification steps, any errors, and how you resolved them.

J. Score Sheet

Criteria	Description	Points
Safe preparation	Correctly identifies and uses a test database; protects credentials and existing data.	10
Transaction rollback	Runs the test and explains the observed result accurately.	20
Backup creation	Creates and identifies a SQL backup file.	15
Restore procedure	Imports backup into a separate test database without overwriting the source.	20
Verification evidence	Compares row counts and sample records, and discusses discrepancies honestly.	15
Scenario analysis	Selects and justifies recovery approaches for different failures.	10
Reflection and documentation	Explains RPO/RTO and submits clear	10

	evidence and conclusions.	
TOTAL		100

K. Troubleshooting Guide

Problem	What to check
CREATE TABLE fails because it exists	Inspect the table first. Reuse your existing lab table or ask your instructor how to reset the lab safely.
ROLLBACK does not remove the row	Confirm the table uses InnoDB and all statements ran in the same connection/transaction.
Export file is empty or missing	Check browser downloads, export success message, selected database, and file size.
Import reports table already exists	Confirm you selected the new empty recovery database, not the original database.
Import fails due to permissions or size	Record the error and ask your instructor for an approved local recovery environment. Do not import over the source database.
Row counts differ	Check whether data changed after backup, whether all tables were exported, and whether import reported errors.

References

- Laravel Documentation — Database: https://laravel.com/docs/12.x/database
- MySQL 8.4 Reference Manual — InnoDB Recovery: https://dev.mysql.com/doc/refman/8.4/en/innodb-recovery.html
- MySQL 8.4 Reference Manual — mysqldump: https://dev.mysql.com/doc/refman/8.4/en/mysqldump.html
- MySQL 8.4 Reference Manual — Backup and Recovery: https://dev.mysql.com/doc/refman/8.4/en/backup-and-recovery.html

Use documentation matching the MySQL version installed in your environment. This worksheet is intended for local/test databases. Never perform destructive recovery experiments on a live system.